

# Labelled-pixel metrics are blind to false positives in sparse crack segmentation: a negative-control criterion for operator-trained models in transmission X-ray microscopy

**DRAFT.** Every number below is read from a recorded artifact in this repository, and the provenance is given inline. Items still outstanding before submission are collected in §8 and marked **[TODO]** where they occur. Author list, affiliations and funding are placeholders.

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## Abstract

Interactive segmentation — a frozen vision-foundation-model encoder, a light classifier, and a human correcting the output — is now a standard way to build a task-specific segmenter from a few dozen images. Model selection in this loop is almost always driven by agreement with the operator's labels: intersection-over-union, AUC, recall at matched false-positive rate. We show on transmission X-ray microscopy (TXM) of fatigue cracks in 316L stainless steel that this practice has a structural blind spot. In a controlled four-arm comparison at a fixed training budget, a gradient-boosted classifier outperformed a neural-network ensemble on intersection-over-union over labelled pixels (0.748 against 0.714) while marking **4.7 times more** area of specimens confirmed to contain no crack (2.141% against 0.451%). The failure is invisible to every labelled-pixel metric by construction, because the material on which it occurs is precisely the material nobody annotates. We propose a mandatory negative-control axis — predicted area on confirmed-empty specimens — as a deployment criterion, and report that it refused two candidate models in this project that labelled-pixel metrics accepted. We additionally report (i) a label-side correction for annotation width bias, in which each hand-drawn stroke is narrowed to the darker core inside it before sampling, reducing output mask half-width from 22 px to 5 px against a measured 2.5–3 px core; and (ii) two measurement pitfalls with consequences beyond this dataset — crack area fraction is not comparable across fields of view, and operators systematically zoom in as cracks grow, so naive growth curves partly measure the camera; and a frozen encoder produced predicted areas differing by a factor of two across compute stacks on bit-identical input. Tool, models, corrections and analysis code are released.

**Keywords:** interactive segmentation, human-in-the-loop, model selection, false positives, transmission X-ray microscopy, fatigue cracks, 316L, reproducibility

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## 1. Introduction

Quantifying a fatigue crack from tomographic imaging requires a segmentation of the crack, and segmentation of thin, sparse, low-contrast features remains the bottleneck. Fully supervised deep learning needs dense annotation that nobody has. The now-conventional alternative is interactive: take a frozen foundation-model encoder, put a small classifier on top, let an operator correct the output, and retrain from those corrections.

That loop has a model-selection problem which, to our knowledge, has not been stated plainly. The operator's corrections are the only labels available, so every metric used to decide whether a retrain is an improvement is computed on the pixels the operator touched. In a sparse task the overwhelming majority of the specimen is untouched — not because it is unimportant, but because there is nothing there to mark. A model can fit the annotated distribution better and behave far worse on the unannotated remainder, and no metric computed over annotations can detect it.

We encountered this concretely. This paper reports:

1. **A controlled demonstration of the blind spot** (§5.1) and a negative-control criterion that catches it, together with an honest account of how often the two axes actually disagree. 2. **A label-side correction for annotation width bias** (§5.2). Brush strokes are far wider than the crack they mark; a model trained on them reproduces the brush. Existing remedies modify the loss or the evaluation metric; we modify the label. 3. **Two measurement pitfalls** (§5.3) that we believe affect published tomographic crack quantification generally. 4. **An application** (§5.4): 71 TXM frames across four 316L specimens, segmented from operator corrections alone.

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## 2. Related work and what is new here

**Interactive segmentation with foundation models.**  $\mu$ SAM [1] adapts Segment Anything to microscopy with interactive correction and fine-tuning. Cellpose 2.0 [2] provides a human-in-the-loop retraining pipeline and reports that 100–200 corrected regions suffice for a custom model. Both are developed for and evaluated on biological imaging.

**Foundation models in materials imaging.** MatSAM [3] extracts eleven classes of metallic microstructure from SAM without training, using an automated point-prompt strategy. SAM-I-Am [4] applies semantic boosting to SAM for atomic-scale electron micrographs. Both are prompt-driven and training-free; neither learns from operator corrections.

**Machine learning on TXM.** CNN segmentation of TXM tomograms is established [5], as are classical machine-learning segmentation pipelines for TXM [6]. These are automatic segmenters trained on curated ground truth, not interactive tools.

**Human-in-the-loop for microstructure.** A unified microstructure segmentation approach using weak supervision and active learning [7] is the closest materials-side analogue.

**Thin-structure annotation quality.** Label error in thin-crack detection has been characterised [8]; topology-aware losses such as cIDice and skeleton-based evaluation metrics address thin-structure segmentation from the loss and metric side respectively.

**What is therefore not new here.** The architecture. A frozen encoder plus a light classifier plus interactive correction is the established pattern, and we claim no novelty for it. Nor is machine learning on TXM new [5,6], nor SAM applied to materials imaging [3,4].

**What we believe is new.** (i) The demonstration that labelled-pixel metrics are structurally blind to false positives in this loop, with a controlled measurement of the effect size and a deployment criterion that catches it. (ii) Correcting annotation width bias at the label rather than the loss or the metric. (iii) The two measurement pitfalls of §5.3. (iv) As a secondary point, the combination — corrections-only training with a negative-control gate, on TXM fatigue cracks — appears not to have been reported; the materials-side SAM work is prompt-driven and the interactive-correction work is biological.

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## 3. Method

### 3.1 Features

Each pixel is described by 273 numbers: 17 isotropic hand-crafted measurements (local darkness, darkness relative to neighbourhood, edge and ridge responses at several scales) and 256 channels from the Segment Anything ViT-H image encoder, used frozen. The encoder runs over 1024 px tiles at a stride of 896 px, producing a 64x64x256 grid per tile; per-pixel values are read from that grid by bilinear interpolation, and overlapping tiles are combined with a Hann window so tile boundaries do not appear in the output.

### 3.2 Classifier

Two multilayer perceptrons —  $17 \rightarrow 64 \rightarrow 32 \rightarrow 1$  and  $273 \rightarrow 128 \rightarrow 64 \rightarrow 1$ , each behind a standard scaler — and the mean of their two probabilities. The ensemble beats either member in all five cross-validation folds (member means 0.726 and 0.786 against 0.811 for the average). No external labels are used at any stage: both members are fitted on the operator's corrections only.

### 3.3 Label narrowing

A brush stroke marking a crack is far wider than the crack. Measured against the darkest fifth of each stroke, strokes on these frames are 2.6x to 15x wider than the feature, with a median half-width of 26.9 px against a 3.2 px dark core. Before sampling training rows, each stroke labelled *crack* is narrowed to the pixels darker than the local mean of their own neighbourhood, and the discarded ring is sampled as background. The narrowing is verified per frame and **declined** where the assumption fails — on 2 of 61 frames the dark core does not survive a 60% retention test, and those frames keep their original labels.

### 3.4 Deployment gate

A retrain is accepted only if it passes both of:

- **Held-out agreement.** Cross-validation grouped by whole image, so train and test never share

a frame, against an absolute floor of 0.60 IoU.

- **Negative control.** Predicted crack area on six specimens the operator has confirmed contain

no crack, against a tolerance of 0.5 percentage points.

A model failing either is written to disk and remains selectable for inspection, but does not become current. Both axes are recorded for every retrain.

## 4. Data

Seventy-one TXM frames, 2.9–32.1 megapixels, float32, of four 316L stainless steel specimens:

| set                   | frames | ordered by | range                         |
|-----------------------|--------|------------|-------------------------------|
| HC 316L, fatigue      | 27     | cycles     | 200 → 1790                    |
| wrought 316L, fatigue | 14     | cycles     | 0 → 1300                      |
| B2                    | 17     | depth      | 333.75 → 343.75 $\mu\text{m}$ |
| B3                    | 13     | depth      | 268.13 → 388.13 $\mu\text{m}$ |

Six further specimens are confirmed by the operator to contain no crack and are reserved entirely for the negative-control axis; they are never trained on. Total operator annotation: approximately 264 million labelled pixels across the 71 frames, from a single annotator.

**[TODO]** Instrument, beam energy, voxel size, reconstruction method, loading conditions and specimen geometry. The absence of a recorded  $\mu\text{m}/\text{px}$  scale is a substantive limitation (§7).

## 5. Results

### 5.1 Labelled-pixel metrics are blind to false positives

**Figure 1** (docs/paper\_figs/fig1\_metric\_blindness.png). Four arms at a fixed 400,000-row training budget, crossing {reference frames included, excluded} with {gradient boosting, MLP ensemble}, each scored on both gate

axes:

| arm                                 | IoU on labelled pixels | crack-free area called crack |
|-------------------------------------|------------------------|------------------------------|
| HGB, no reference frames            | <b>0.748</b>           | 2.141%                       |
| MLP ensemble, no reference frames   | 0.714                  | <b>0.451%</b>                |
| HGB, with reference frames          | 0.869 *                | 2.540%                       |
| MLP ensemble, with reference frames | 0.921 *                | 0.544%                       |

\\* trains on the frames it is scored on; reported for completeness only.

On the uncontaminated pair, the classifier that wins on labelled pixels is **4.7x worse** on material confirmed to contain no crack. The effect persists in both training compositions, so it is a property of the classifier rather than of the data mixture; removing the reference frames slightly *lowers* false positives (0.544% → 0.451%).

Every metric that favoured the losing choice — IoU, AUC, recall at matched false-positive rate, cross-group AUC — is computed over labelled pixels. Crack-free specimen is exactly the material nobody labels: there is nothing to find, so nothing is marked. A model that separates the labelled distribution better while behaving worse off it is therefore undetectable from annotations alone, in principle and not merely in practice.

**How often do the axes disagree?** Figure 2 places the five recipe-tagged models actually deployed in this project on both axes. Kendall  $\tau$  between the two rankings is **+0.80**: nine of ten pairs concordant. One inversion: `corrections_only_v4_overlap` raised IoU from 0.776 to 0.789 while worsening crack-free false positives from 0.188% to 0.209%.

We therefore make the weaker and defensible claim: labelled-pixel agreement is usually informative, is **never sufficient**, and provides no signal about whether this failure mode has occurred. In this project the negative-control axis refused two candidate models that labelled-pixel metrics accepted — one at 2.141% and an earlier one at 22.4% predicted area on crack-free material.

## 5.2 Annotation width bias, corrected at the label

**Figure 3.** Narrowing each stroke to its dark core before sampling, against training on full strokes:

|                               | full-stroke labels | narrowed labels |
|-------------------------------|--------------------|-----------------|
| held-out grouped IoU          | 0.789 ± 0.040      | 0.811 ± 0.023   |
| precision / recall            | 0.933 / 0.837      | 0.936 / 0.860   |
| crack-free area called crack  | 0.209%             | 0.174%          |
| <b>output mask half-width</b> | <b>22 px</b>       | <b>5 px</b>     |

The mask stops reproducing the brush: half-width falls from 22 px to 5 px against a measured 2.5–3 px dark core, so roughly half the remaining gap is closed and the residual is honestly reported rather than claimed away.

**A comparison we cannot make cleanly.** The two IoU values are measured against *different targets* — the narrowed recipe is scored against narrowed labels — so that row is not like-for-like and we do not rest any claim on it. The comparable axes are mask width and crack-free false positives; both improve. **[TODO]** A matched-target ablation, training on both label variants and evaluating both against a single held-out label set, is outstanding and should be completed before submission.

Existing remedies for thin-structure annotation act on the loss (cDice and related topology-aware objectives) or on the metric (skeleton-based evaluation). The correction here acts on the label, costs one filter pass, and is independent of both model and loss.

## 5.3 Two measurement pitfalls

**(a) Crack area fraction is not comparable across fields of view.** Crack area fraction is crack pixels divided by frame pixels, so it depends on how much surrounding material is in shot. Three frame pairs in this dataset image the same physical state at two fields of view:

| state                    | wide field       | close field     |
|--------------------------|------------------|-----------------|
| wrought, 1100 cycles     | 2.03% (10.4 MP)  | 6.25% (8.8 MP)  |
| B2, 343.75 $\mu\text{m}$ | 12.29% (23.5 MP) | 23.13% (4.0 MP) |
| B3, 388.13 $\mu\text{m}$ | 9.04% (23.3 MP)  | 10.61% (3.9 MP) |

Nothing about the specimen changed. Absolute pixel counts do not rescue the comparison: the 23.5 MP B2 frame contains 2,888,587 crack pixels against 924,887 in the 4.0 MP frame, because a wider field simply contains more crack.

**And the operator zooms in as the crack grows.** Correlation between fatigue cycles and frame megapixels is  $-0.76$  (HC) and  $-0.68$  (wrought). A growth curve plotted from area fraction therefore rises partly because the crack grew and partly because the frame shrank.

**Shape descriptors are nearly immune.** Correlation with frame megapixels, over all 64 frames containing crack: mean width  $-0.05$ , branch ratio  $+0.10$ , skeleton fractal dimension  $-0.10$ , branch points per unit length  $-0.06$  — against  $-0.49$  for area fraction. The recommendation follows: report field-of-view-independent descriptors, or hold the field constant.

**(b) A frozen encoder is not automatically reproducible across compute stacks. Figure 4.** Identical code, bit-identical decoded input (verified by checksum), identical model files: predicted area 0.1880 on four environments and **0.0925** on one, with the mask shattering from 968 connected components to 21,019 and the probability mean falling from 0.381 to 0.369. A two-arm matrix on the outlying machine isolated the cause to the encoder's compute device; four alternatives (image decode, BLAS backend, tiling geometry, image-processor backend) were tested and refuted. Anyone publishing numbers from a frozen foundation-model encoder should pin and report the device.

## 5.4 Application: crack development in four 316L specimens

Analysis is restricted to within-field-of-view comparisons throughout, per §5.3.

**Growth, unconfounded.** Six wrought-316L frames at a constant 21.9–22.2 MP field: no crack at 300 cycles, 39,313 crack pixels at 800, 158,310 at 1000, 447,473 at 1100. Growth rate rises roughly **37x** between the 800→900 and 1000→1100 intervals.

**Initiation.** Two HC frames at a matched ~32 MP field bracket initiation: 0 crack pixels at 400 cycles, 234,343 (0.731% of frame) at 600. At first detection the crack is already a single region holding 95.4% of the crack area.

**Depth.** Six B3 frames at a constant 3.9–4.0 MP field: 2.79% → 10.61% over 380.00 → 388.13  $\mu\text{m}$ , a 3.8x increase across 8  $\mu\text{m}$ , remaining a single connected region throughout.

**Morphology develops in roughness and branching, not in meander.** Skeleton fractal dimension rises with crack development in all four sets ( $r = +0.56, +0.66, +0.54, +0.59$ ) and branch ratio rises strongly in three ( $+0.49, +0.72, +0.72$ ), while main-path tortuosity stays flat. Every one of these descriptors is field-of-view independent per §5.3.

**One dominant crack, not a network.** Per-region analysis of 439 regions shows the largest region holds a median 79% (HC), 80–99% (wrought) and 83–100% (B2) of each frame's crack area. Region *counts* are not crack counts: the export retains elongated components down to a 200 px floor, so a frame reporting 16 regions had 9 of them below 2,000 px holding 0.59% of the area. Frame-level means over regions are therefore contaminated — per-region mean width and boundary roughness correlate  $+0.77$  and  $+0.72$  with log region area — and morphology should be read off the dominant region only.

**What we do not claim.** Nothing here distinguishes transgranular from intergranular cracking. That is a statement about the crack path relative to grain boundaries, and no grain-boundary information is registered to these frames. We report straight-segment lengths (median 53–60 px) and turn angles (median 25–27°), strikingly consistent across three specimens, as a fingerprint that becomes diagnostic only once a grain size and a  $\mu\text{m}/\text{px}$  scale are supplied.

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## 6. Discussion

The negative-control axis is not a second opinion on the same evidence. It is the only axis that reads material the training distribution does not cover, which is why it can contradict every other metric at once. Its cost is low — a handful of specimens the operator confirms are empty, never trained on — and in this project it twice prevented a regression that labelled-pixel metrics endorsed.

The generalisation is not specific to cracks. Any task in which the positive class is sparse, thin, and annotated by a human who marks only what they find will have the same structure: vessels, fibres, road networks, weld defects, delamination. Wherever the negative class is *unlabelled rather than labelled-negative*, agreement with annotation cannot bound false positives.

Label narrowing addresses a different failure of the same asymmetry. The operator's stroke encodes *where* a crack is far more reliably than *how wide* it is, and a model trained on the stroke inherits the width. Correcting the label rather than the loss keeps the fix independent of architecture.

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## 7. Limitations

- **No physical scale.** No  $\mu\text{m}/\text{px}$  is recorded in the exported data, so all lengths and areas here are in pixels or as a fraction of frame. Crack length, opening and growth rate in physical units are unavailable, as is any comparison to  $\Delta K$ . This is the single most consequential limitation and it is a recording gap, not an analysis one.
  - **Two-dimensional.** These are areas on single slices, not crack volumes or crack-front shapes.  
A crack growing in depth and one growing in width both read as more area.
  - **One annotator, no repeats.** No inter- or intra-operator uncertainty can be estimated, so no error bars appear on any measurement in §5.4.
  - **The masks are heavily human-curated.** Across the four sets the operator painted 4.6, 9.8, 9.3 and 7.6 million pixels as crack against final mask areas of 6.3, 9.4, 10.6 and 7.3 million. These numbers reflect one person's judgement as much as the model's.
  - **The width comparison in §5.2 is not matched-target** — see the [TODO] there.
  - **Six negative controls** is a small sample for the axis on which the paper's main claim rests.
  - **A data-integrity defect** remains in the B3 set: two frames labelled 380.00  $\mu\text{m}$  in the same field-of-view tier disagree by 2x in area and 4.7x in skeleton length.
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## 8. Outstanding before submission

1. Matched-target ablation for §5.2. 2.  $\mu\text{m}/\text{px}$  scale from the instrument headers; re-express §5.4 in physical units. 3. Second annotator on  $\geq 10$  frames, for uncertainty. 4. Instrument and loading metadata for §4. 5. Resolve the duplicated 380.00  $\mu\text{m}$  frames. 6. Decide figure/supplementary split — 36 per-set charts exist and most belong in

supplementary.

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## Data and code availability

Tool, both shipped models, the analysis code and all 71 operator correction masks are in the project repository. Continuous integration runs the install, the self-test suite and a real ensemble prediction on Linux and macOS on every commit. **[TODO]** Deposit the corrections and models with a DOI; they are the irreplaceable artifact and the models are reproducible from them.

## References

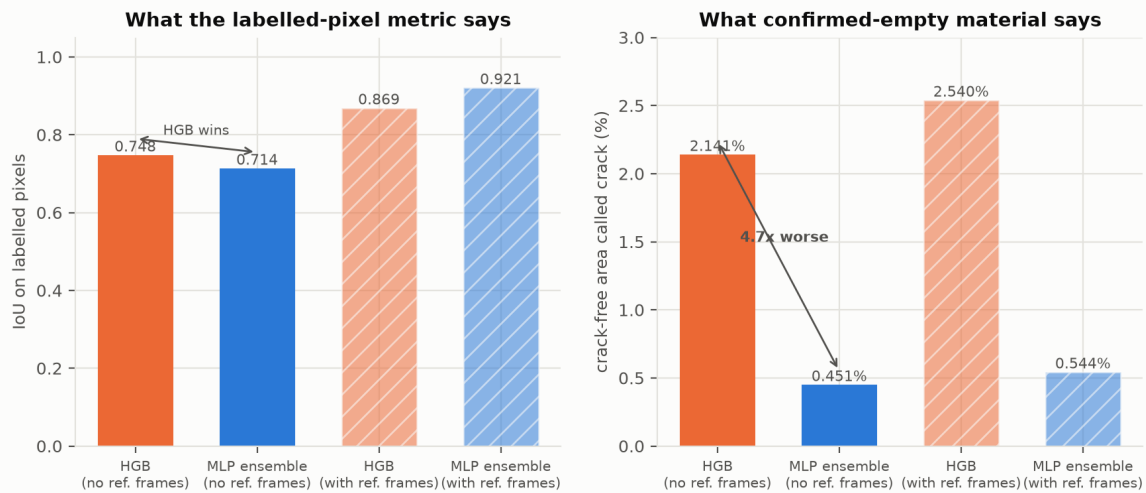
[1] Archit et al. Segment Anything for Microscopy. *Nature Methods*, 2024. [2] Pachitariu & Stringer. Cellpose 2.0: how to train your own model. *Nature Methods*, 2022. [3] MatSAM: a training-free approach to extracting material microstructures via a visual large model. *Acta Materialia*, 2025. [4] SAM-I-Am: semantic boosting for zero-shot atomic-scale electron micrograph segmentation. *Computational Materials Science*, 2024. [5] Automated correlative segmentation of large transmission X-ray microscopy tomograms using deep learning. *Materials Characterization*, 2018. [6] Machine-learning-based algorithms for automated image segmentation of transmission X-ray microscopy. *JOM*, 2021. [7] A unified microstructure segmentation approach via human-in-the-loop machine learning. *Acta Materialia*, 2023. [8] Xu et al. How do label errors affect thin crack detection by DNNs. *CVPR Workshops*, 2023.

**[TODO]** Complete bibliographic details, volumes, pages and DOIs.

## Charts

fig1\_metric\_blindness.png

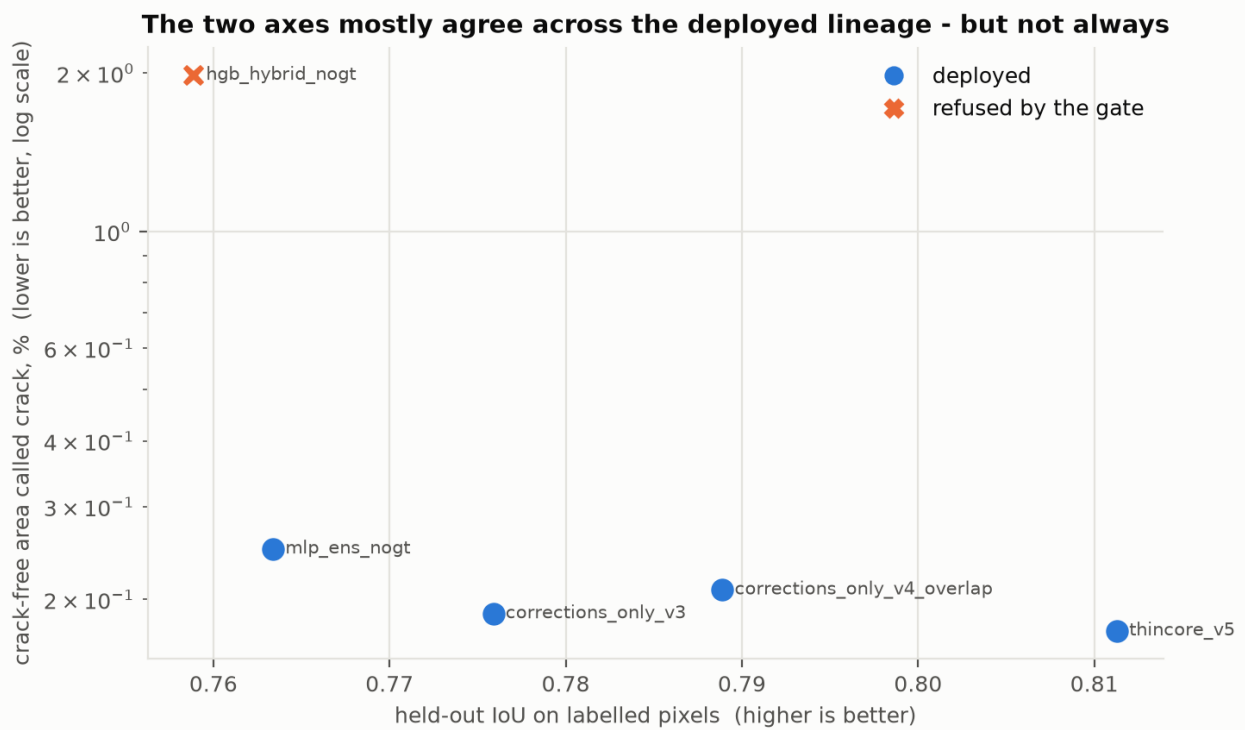
**The classifier that wins on labelled pixels is 4.7x worse on material known to contain no crack**



Four arms at a fixed 400,000-row budget (research/fp\_attribution.json). Hatched bars train on the frames they are scored on and are shown only for completeness. Blue = MLP ensemble, orange = HGB.



fig2\_lineage\_two\_axes.png



Five recipe-tagged models, each measured on its own corrections only. Kendall tau between the two rankings is +0.80: 9 of 10 pairs concordant. The exception is `corrections_only_v4_overlap`, which raised IoU 0.776 to 0.789 while worsening crack-free false positives 0.188% to 0.209%. So IoU is usually informative and never sufficient.

fig3\_label\_narrowing.png

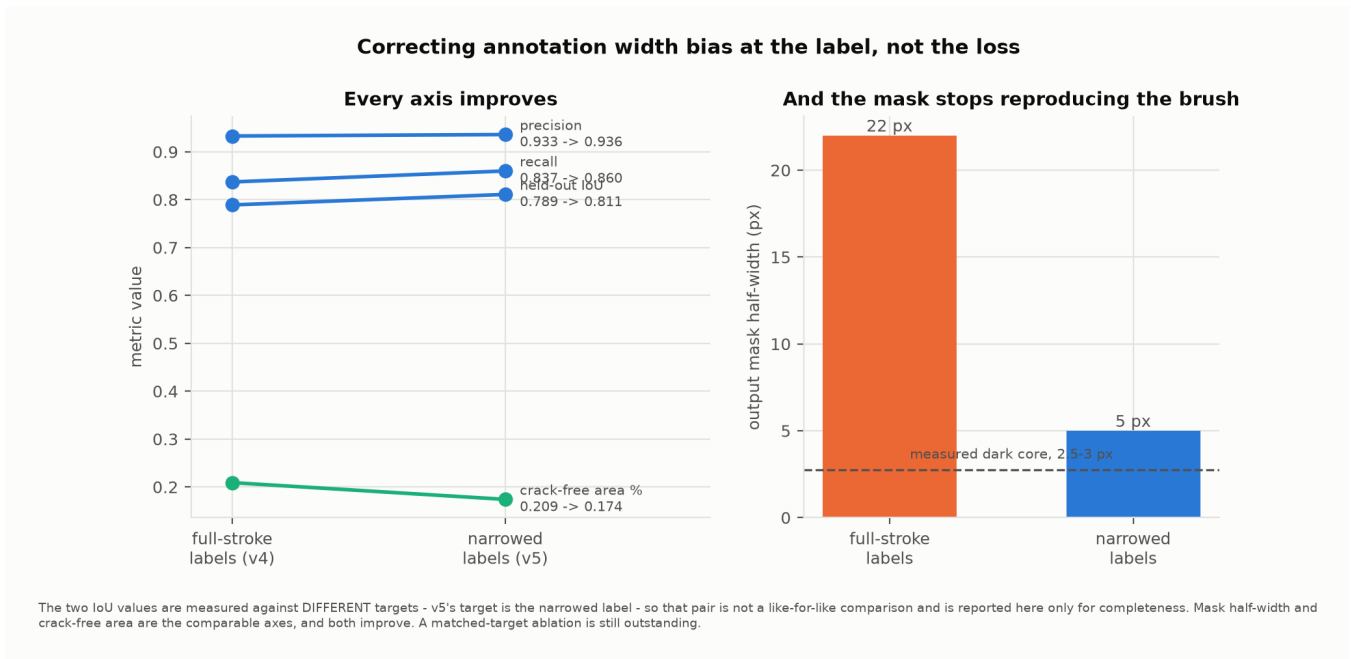
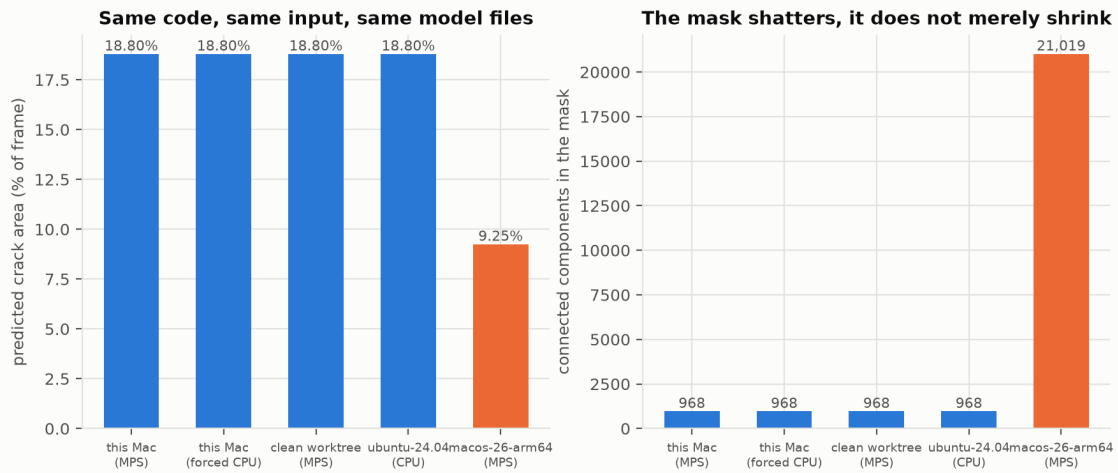


fig4\_device\_reproducibility.png

**A frozen encoder is not automatically reproducible across compute stacks**



Input verified bit-identical across all five (same sha1 of the decoded frame). A two-arm matrix on the outlying runner image isolated the cause: forced to CPU it reproduces 0.1880 and 968 components. Decode, BLAS, tiling geometry and the image processor backend were each tested and refuted.